ALICE L WALTON SCHOOL OF MEDICINE

Management of Chronic Heart Failure

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Adapted from: Management of Chronic Heart Failure

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Describe chronic heart failure (CHF) and how the body responds.
2. Outline the classes of drugs used to treat chronic systolic heart failure and their mechanisms of action.
3. Outline a treatment plan for chronic heart failure, both systolic and diastolic.

CASE CONNECTION

MR, a 64-year-old man, presents at his primary care clinic with progressive shortness of breath and ankle swelling over three months. On exam he has stable vital signs, bilateral rales, S3 heart sound and 2+ bilateral lower extremity pitting edema. He reports difficulty exercising and waking at night due to breathlessness. AB is a third-year medical student reviewing MR's symptoms during an internal medicine rotation and wonders about the underlying pathophysiology and treatment options for chronic heart failure. What processes drive MR's symptoms? Which medication classes would you consider in his management? Consider your answers as you read, and we'll revisit MR at the end of the brick.

GO TO CONCLUSION ↓

What Is Heart Failure? (LO#1)

Heart failure (HF) is a syndrome in which the heart is unable to pump sufficient blood to meet the needs of the body. This may be because of a weakened myocardial pump with low contractility (systolic dysfunction) or a thickened myocardium due to left ventricular hypertrophy (LVH), with impaired filling (diastolic dysfunction, or nonsystolic HF). As previously mentioned, heart failure is subcategorized into HFrEF, HFpEF and HFmrEF. In addition to these categories, heart failure can present as acute decompensated systolic heart failure and acute decompensated diastolic heart failure. Patients can also have acute c [illegible] and diastolic heart failure.

Body's Response to Heart Failure

In the short-term, the body responds to heart failure in two ways:

- It activates the sympathetic nervous system to increase heart rate, contractility, and arteriole vasoconstriction.
- It activates the renin-angiotensin-aldosterone system (RAAS) to increase intravascular pressure through angiotensin-mediated vasoconstriction and aldosterone-mediated fluid retention.

Together these responses to HF increase afterload (via vasoconstriction) and increase preload (via increased intravascular volume) in an attempt to maintain tissue perfusion (effective intravascular volume).

Although there is no one-size-fits-all treatment for heart failure, we can give drugs to **reduce both preload and afterload** and **reduce cardiac remodeling.** By reducing all three, we can slow the progression or even reverse disease, relieve symptoms, and reduce complications.

Treating Heart Failure

How then do we treat HF? The best treatment depends on many factors. Recall that HF can be either systolic or diastolic, and systolic and diastolic HF have different causes and pathophysiology. It makes sense that systolic and diastolic HF would have different treatments as well.

In this brick we will mostly focus on the management of chronic systolic HF. We'll say a little bit about diastolic HF at the end as well as acute decompensated hear [illegible] ure is different than a patient with chronic heart failure, primarily in the order in which medications are started.

What are the primary goals for the drugs given to reduce chronic heart failure?

What Are the Different Classes of Drugs Used to Treat Heart Failure? (LO#2)

Knowing the mechanisms behind each drug helps us understand the rationale behind their benefits and so guides us on when to use them. To that end, the drugs have been grouped based on their ultimate cardiovascular effect.

Basic Strategies for Treating Heart Failure

Many of the drugs are covered thoroughly in their own bricks, so we will only summarize those drug [illegible] the mechanisms before we learn how t [illegible]

1. **Decrease preload:** Although you might think that increasing preload would help the cardiac force of contraction, it also causes two of the major symptoms of HF: pulmonary and peripheral edema. So, preload reduction is a mainstay of HF therapy since it reduces the patient's symptoms.
2. **Decrease afterload:** This increases the cardiac output for any given level of contractility, increasing tissue perfusion. It also reduces the work of contraction. These drugs are used in all patients with HF.
3. **Increase contractility:** Inotropic drugs increase the force of contraction directly, increasing the cardiac output. However, this comes at the risk of increasing the work required of the damaged myocardium and, in many cases, increasing the risk of arrythmias. The inotropic drugs **do not** improve mortality, despite the "logic" of their mechanism, so are not used routinely unless other meds are not helping or in the special case of cardiogenic shock.
4. **Decrease myocardial remodeling:** Remodeling is the alteration of cardiac structure in response to cardiac injury. This remodeling is a major cause of morbidity and mortality from HF. So, its reduction is a prime goal of HF therapy.

In practice, we strive to do #1, #2, and #4 to improve symptoms and mortality, and only add #3 when other strategies fail. So now let's look at the various categories of drugs used to manage heart failure (figure1).

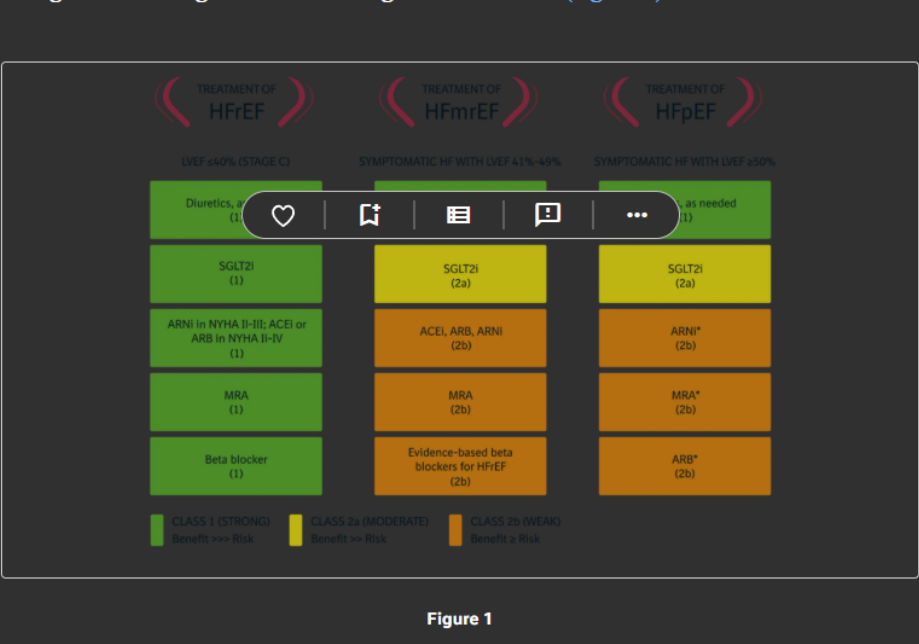


Figure 1

GDMT (Guideline-Directed Medical Therapy)

GDMT (Guideline-Directed Medical Therapy) refers to a standardized approach for treating heart failure, particularly heart failure with reduced ejection fraction (HFrEF). It typically includes a combination of four key medication classes: RAS Inhibitors (ACEI, ARB, ARNI), Beta Blockers (BB), MRAs (Aldosterone Antagonists) and Sodium-Glucose Cotransporter (SGLT2) Inhibitors. When thinking about what medications to add for chronic systolic heart failure, it is important to focus these medications because they decrease mortality.

RAS Inhibitors- ACEI, ARB, ARNI

Angiotensin-Converting Enzyme Inhibitors. Angiotensin-converting enzyme (ACE) inhibitors end in the suffix -pril (eg, lisinopril, enalapril, benazepril). A close cousin to ACE inhibitors are **angiotensin-receptor blockers** [ARBs]). They generally end in the suffix -sartan (eg, losartan, valsartan).

These drugs reduce the effects of angiotensin II (AT-II). This helps in HF in two ways. AT-II is a vasoconstrictor, so blocking it will reduce systemic vascular resistance (SVR) and therefore **reduce afterload**. AT-II also stimulates aldosterone synthesis, which causes renal sodium reabsorption (so does AT-II, directly). Blocking these effects will **decrease** renal sodium and water reabsorption, intravascular volume, and **preload.**

Last, but not least, ACE inhibitors and ARBs **decrease cardiac remodeling, improving prognosis and mortality for the long term.**

The major adverse effects of ACE inhibitors are bradykinin-mediated cough and hyperkalemia. Less common are acute kidney injury in those with bilateral renal artery stenosis and (rarely) angioedema. The incidence of these adverse effects is greatly reduced in ARBs.

ARNI (angiotensin receptor blocker/neprilysin inhibitor) medications A new combination drug, **sacubitril/valsartan** inhibits the degradation of natriuretic peptides by neprilysin. While technically not diuretics, these drugs have similar actions to diuretics. Natriuretic peptides are released by the atria and ventricles when stretched, and they cause the kidney to excrete sodium (and water), thereby **reducing** intravascular volume **(preload)**. By inhibiting neprilysin, sacubitril increases natriuretic peptide levels and allows for more fluid excretion. This drug decreases mortality in a subset of patients with systolic heart failure but can cause hypotension.

β-Blockers. Only three β-blockers have been demonstrated to be safe and effective in reducing morbidity and mortality in patients with heart failure: **metoprolol succinate, carvedilol, and bisoprolol.**

CLINICAL CORRELATION

Remember way back to the Foundations of Pharmacology course when we discussed β-blockers and which specific receptors they have higher affinity for. You might recall that we would classify metoprolol succinate and bisoprolol as 'cardioselective' because of their high affinity for β_1 receptors and very low affinity for β_2 receptors. Carvedilol is unique because it has good selectivity for $\beta_1 > \beta_2$ receptors, as well as alpha-1 receptors. This makes carvedilol a great choice for patients with heart failure and concomitant hypertension.

From physiology, we would predict that β-blockers, specifically because of their activity on β_1 receptors, would affect the heart by reducing contractility and lowering the heart rate. Why would we want to do either of these actions in CHF, since these effects would both be expected to decrease cardiac output?

In patients with acute, decompensated HF and hypotension we do avoid these drugs initially for these exact reasons. But in chronic HF treatment we use them along with ACE inhibitors. Why? First, because β-blockers also block β_1 receptors on juxtaglomerular cells in the kidney. Activation of these receptors normally stimulates renin secretion, so, conversely, blocking them reduces renin secretion and blocks the RAAS, rather like the ACE inhibitors do (not to the same extent, though). **They will also reduce preload and afterload.**

Other benefits of β-blockers are that they block the persistent overstimulation of the heart by the sympathetic nervous system (termed neurohormonal activation) and can also **"reverse remodel"** the cardiac remodeling that can chronically weaken th [illegible] gain back some of its pumping function. [illegible] short-term reduction in cardiac output that might be seen.

β-Blockers should be avoided in patients with cardiogenic shock (severe acute heart failure), where you do not want to decrease cardiac output acutely. They should also be avoided in patients with active asthma or other causes of bronchospasm, which they can worsen. They may cause erectile dysfunction in some males.

SGLT–2 Inhibitors

SGLT-2 inhibitors (e.g., dapagliflozin and empagliflozin), a mainstay of treatment for diabetes, are now foundational therapies for patients with heart failure, regardless of diabetes status. By inhibiting sodium–glucose cotransporter-2 in the proximal tubule of the kidney, these agents increase urinary glucose and sodium excretion, resulting in mild natriuresis (osmotic diuresis), **reduced preload and afterload**, and favorable effects on cardiac metabolism and remodeling. Unlike traditional diuretics, SGLT-2 inhibitors provide significant reductions in heart failure hospitalizations and mortality. Common considerations include volume depletion, genitourinary infections, and rare risks such as euglycemic diabetic ketoacidosis in susceptible patients. Mechanisms, adverse effects, and renal considerations for this drug class will be discussed in greater detail in the Endocrine course.

What are the beneficial effects of ACE inhibitors and ARBs in heart failure?

Loop Diuretics

Loop Diuretics (e.g., furosemide, torsemide, and bumetanide) are essential for most patients with HF since they increase urine sodium and water losses, reducing intravascular volume and preload. Diuretics, however, **do not decrease mortality.** They are only beneficial for symptom management. Loop diuretics, i.e. mechanism, ADRs, etc. will be discussed in more detail in the Renal course.

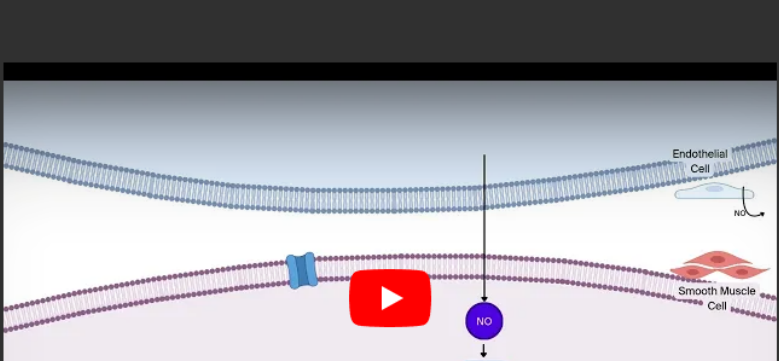
Spironolactone

Spironolactone, a mineralocorticoid receptor antagonist (MRA), also commonly classified as a potassium-sparing diuretic, and is recommended for select patients with HFrEF because it blocks aldosterone-mediated sodium and water retention, myocardial fibrosis, and ventricular remodeling. By antagonizing aldosterone, spironolactone provides both symptomatic benefit and a proven reduction in morbidity and mortality when used in appropriate patients already receiving guideline-directed medical therapy. It is often recommended in patients who have HFrEF with continued symptoms despite adding ACE vs ARB vs ARNI, BB and [illegible] ed for those with NYHA III or IV Heart failure (look at end of brick for classification) and an LVEF of < 35%.

Key considerations include its potassium-sparing effects and risk of hyperkalemia, as well as endocrine-related adverse effects such as gynecomastia. **Eplerenone** is another MRA without the risk of gynecomastia in males. The detailed mechanism, adverse effects, and additional clinical considerations for spironolactone will be covered more extensively in the Renal course.

Nitrates

Nitroglycerin is a potent venodilator and allows veins to hold extra blood (increased vessel capacity). This means less blood enters the heart, thereby reducing preload with no direct negative inotropic effects. Recall, from the Pharmacology course the mechanism of nitrates.



Nitrates are used in combination with hydralazine as second-line drugs for patients who are unresponsive to ACE inhibitors and/or β-blockers.

Nitrates may cause headache, dizziness, and tachycardia.

What are the most common diuretics used in HF?

Arterial Dilators

These second- or third- line agents are mostly for those who do not respond to ACE inhibitors or ARBs. They directly reduce arteriolar resistance, lowering systemic vascular resistance (SVR) and afterload, however, they do not have any effects on the RAAS.

Hydralazine. Using a [illegible] s directly to relax arterioles and is frequently used with nitrates. The combination of a nitrate and hydralazine will reduce both preload and afterload, respectively. Hydralazine may cause reflex tachycardia and dizziness and, occasionally, drug-induced lupus.

Amlodipine and Felodipine. These are DHP calcium-channel blockers (CCBs) approved for use in systolic HF. They are second-line drugs (along with hydralazine/nitrates). They dilate arterioles, so they decrease afterload. Mechanistically, decreased Ca^{2+} inside arteriolar smooth muscle leads to smooth muscle relaxation, which dilates the arterioles, reducing SVR.

CLINICAL CORRELATION

What about non-DHP CCBs like diltiazem and verapamil for heart failure? Generally, they are not used in the treatment of heart failure. Similar to beta-blockers, they will reduce rate and contractility, however, they do NOT have any effects on the RAAS. This means they do not have the same 'reverse remodel' effects that beta-blockers have, and so they are not used for chronic management.

How do aml [illegible] nts with HF?

Inotropes: Drugs That Increase Contractility

For treating chronic systolic HF, why don't we just take the most direct path and treat the poorly contractile heart with drugs that improve cardiac contractility, i.e. inotropes? Unfortunately, due to toxicity and lack of long-term demonstrated benefit, we generally only use these inotropes in patients with acute decompensated heart failure, cardiogenic shock, or if a patient is unable to tolerate other GDMT drugs. Note that intravenous inotropes (e.g. dobutamine and dopamine) are only used for acute decompensated HF and cardiogenic shock.

Digoxin. Digoxin increases cardiac contractility. It often provides symptomatic relief but cannot prevent or reverse the cardiac remodeling that worsens heart failure. **It does not improve mortality in HF.** Since it has many adverse effects and has no benefit on long-term survival, digoxin is only used in chronic HF when other drugs fail.

Nevertheless, the mechanism is commonly tested on exams, so you should review it. Digoxin increases intracellular calcium, acting via **inhibition of the Na^+/K^+-ATPase** that is seen on all cell types, including cardiac myocytes. Digoxin acts through the following mechanism in cardiac myocytes (Figure 2):

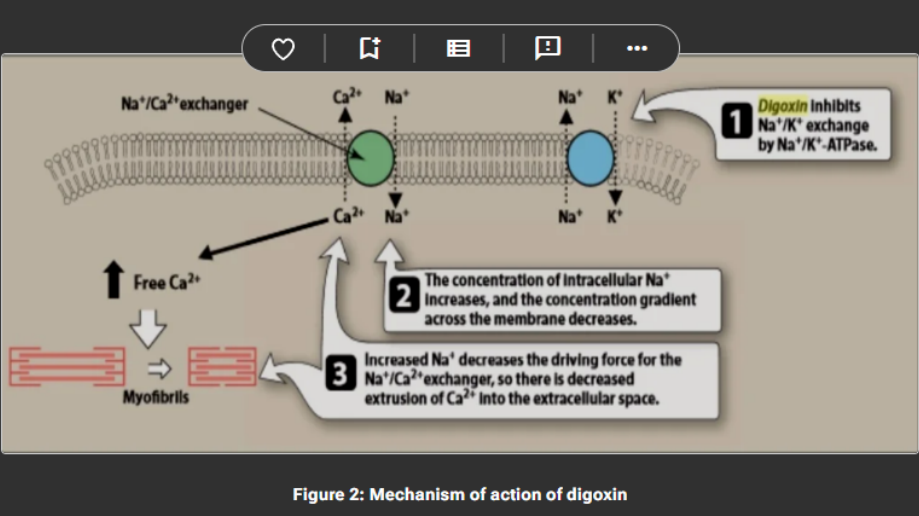

Na⁺/Ca²⁺ exchanger
Ca²⁺
Na⁺
K⁺
Free Ca²⁺
Myofibrils
1 Digoxin inhibits Na⁺/K⁺ exchange by Na⁺/K⁺-ATPase.
2 The concentration of intracellular Na⁺ increases, and the concentration gradient across the membrane decreases.
3 Increased Na⁺ decreases the driving force for the Na⁺/Ca²⁺ exchanger, so there is decreased extrusion of Ca²⁺ into the extracellular space.

Figure 2: Mechanism of action of digoxin

Digoxin has many adverse effects, especially cardiac arrhythmias, that we really want to avoid. Digoxin can induce heart block, bradycardia, ventricular tachycardia, and supraventricular tachycardias. These effects most likely occur if the patient is even slightly hypokalemic, so patients **must maintain a completely normal serum potassium**, often a challenge when they are also taking diuretics.

Other significant adverse effects are hyperkalemia and vision that is decreased and takes on a yellow hue. Digoxin toxicity is often tested on exams.

Why is digoxin not a first-line agent for chronic HF?

Putting It All Together: How Do We Manage Chronic Systolic Heart Failure? (LO#3)

As you've seen, there are numerous drug therapies available to treat chronic systolic HF. Table 1 summarizes these.

Table 1 Drugs used to manage chronic systolic heart failure

Class	General mechanism	Cardiovascular effect	Adverse effects
ACE inhibitors (benazepril, enalapril) ARBs (valsartan, candesartan)	Block RAAS system Reduces angiotensin II level or effect	↓ Preload ↓ Afterload Prevent remodeling	Cough, angioedema, orthostatic hypotension, hyperkalemia
Beta Blockers- Metoprolol, carvedilol, bisoprolol	Block the β-receptor at juxtamedullary cells, decreasing renin release	↓ Preload ↓ Afterload Prevent remodeling	Increased oxygen requirement, fatigue, impaired exercise tolerance, erectile dysfunction
ARNI- Sacubitril/valsartan	Neprilysin inhibitor/ARB; blocks b [illegible] angiotensin II receptor binding	↓ Preload ↓ Afterload	Hypotension, hyperkalemia, cough, angioedema
MRA - Spironolactone	Inhibit aldosterone at receptor	↓ Preload	Gynecomastia, agranulocytosis
SGLT-2 inhibitors	Increase glucose and sodium excretion	↓ Preload ↓ Afterload	Volume depletion, genitourinary infections
Nitrates	Dilate veins, increasing their capacitance Nitrates ↑ NO → ↑ myosin dephosphorylation	↓ Preload	Headache, tachycardia, hypotension, nausea
Loop Diuretics	Increase urine sodium and water losses	↓ Intravascular volume ↓ Preload	
Hydralazine, CCBs	Dilate arteries, decreasing vascular resistance Hydralazine → direct arteriodilation CCBs block Ca^{2+} channels	↓ Afterload	Headache, vomiting, angina, tachycardia, lupus-like syndrome
Digoxin	Blocks Na^+/K^+-ATPase activity	↑ Contractility	Fatigue, nausea/vomiting, dream disturbance, 2nd- /3rd-degree heart block, arrhythmia

In treating chronic HF, we want to maximize cardiac function over the long-term. The goal is to reduce both long-term morbidity and mortality.

First, all HF patients benefit from a low sodium diet (DASH diet) and careful control of blood pres [illegible] 0 mmHg). Most patients will require antihypertensive medications.

Management With Heart Failure Drugs

For systolic HF: Block 1: First start an ACE inhibitor/ARB or ARNI and a β-blocker. They reduce both morbidity and mortality. Block 2: Once the ACE vs ARB vs ARNI and BB are tolerated add the SGLT-2 inhibitor. Block 3: If the patient has a LVEF < 35%, it is also recommended to add an MRA like Spironolactone. If the patient is symptomatic, use a loop diuretic, especially if there is pulmonary edema, or symptomatic peripheral edema that does not resolve with the initial drugs.

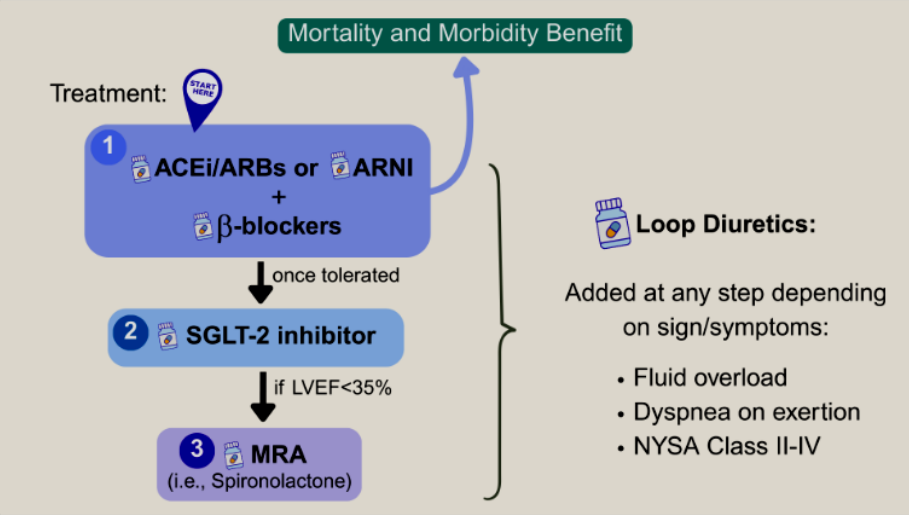

Mortality and Morbidity Benefit
Treatment:
1 ACEi/ARBs or ARNI + β-blockers
once tolerated
2 SGLT-2 inhibitor
if LVEF<35%
3 MRA (i.e., Spironolactone)
Loop Diuretics:
Added at any step depending on sign/symptoms:
• Fluid overload
• Dyspnea on exertion
• NYSA Class II-IV

Figure 3.

If these drugs described in Figure 3 fail, or are not tolerated, try hydralazine/nitroglycerin or digoxin. If the patient's LVEF was above 35% and the ACE vs ARB vs ARNI + BB + SGLT2 was started as well as a loop diuretic, but the patient was still symptomatic, add spironolactone.

Management With Heart Failure Devices

Some patients with severe HF will require additional management beyond just medications.

Implantable Cardioverter-Defibrillator. Patients with systolic HF who survive a cardiac arrest, with very low ejection fraction, or who have ventricular tachycardia are typically treated with an implantable cardioverter-defibrillator (ICD) therapy because of high risk of ventricular arrhythmias, which may be lethal. Typically, a patient has to be maximized on goal director medical therapy at max tolerated doses for 3 months before qualifying to receive an ICD.

Biventricular Pacemaker + ICD Patients with chronic systolic heart failure and prolonged QRS complex (>120 msec) commonly seen in patients with a LBBB, should be evaluated for a bi-ventricular pacemaker _ ICD. These types of pacemakers resynchronize the ventricles to beat together, ultimately improving cardiac output and decreasing heart failure symptoms. A Biventricular ICD is one type of cardiac resynchronization therapy (CRT-D) which is a pacemaker plus ICD.

What drugs used in chronic systolic HF (CHF) have been shown to reduce morbidity and mortality?

What About Diastolic Heart Failure?

We've consistently been focusing in this brick on chronic systolic HF management. A different approach is used in those with left ventricular hypertrophy (LVH) and diastolic HF (also called nonsystolic HF and HF with preserved ejection fraction). In contrast to systolic HF, here no drugs have been shown to reduce cardiovascular remodeling or mortality. Therapy includes:

- Manage hypertension aggressively, with blood pressure reduced to <130/80 mmHg. This can sometimes lead to regression of the LVH. The recommended antihypertensive drugs are ACE inhibitors, ARBs, CCBs, and/or thiazide diuretics.
- Avoid nitrates, β-blockers, and digoxin since they may worsen the outflow obstruction from the thickened ventricle.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to CM [illegible] ribed when he left the hospital?

Lisinopril, an ACE inhibitor, relaxes the blood vessels and reduces the body's hold on sodium and water. This helps the heart work more efficiently. His metoprolol, a β-blocker, works in a similar way and also blocks the activation of the sympathetic nervous system, putting less stress on the heart. Because of CM's volume overload on admission, furosemide, a diuretic, will help his kidneys excrete sodium and water. Other medications may be added now that CM is an outpatient.

Summary

- Heart failure (HF) is a syndrome where the heart cannot pump enough blood to meet the body's needs, categorized into HFrEF, HFpEF, and HFmEF.
- The body's short-term response to HF includes activating the sympathetic nervous system and the renin-angiotensin-aldosterone system (RAAS) to maintain tissue perfusion.
- Treatment goals for chronic HF include reducing preload and afterload, preventing cardiac remodeling, and improving symptoms and mortality.
- Guideline-Directed Medical Therapy (GDMT) for HFrEF includes ACEi, ARBs, beta-blockers, aldosterone antagonists, and SGLT2 inhibitors, which decrease mortality.
- First-line drugs for chronic systolic HF are ACE inhibitors (or ARBs) and beta-blockers, with loop diuretics added for edema management.
- No drug therapies reduce mortality in diastolic HF; management focuses on excellent blood pressure control.

Review Questions

1. Which of the following is not an important mechanism by which angiotensin-converting-enzyme inhibitors and angiotensin-receptor blockers improve symptoms of systolic heart failure?

A. ● Increasing inotropy
B. ○ Preventing cardiac remodeling
C. ○ Preventing fluid retention
D. ○ Reducing afterload
E. ○ Reducing preload

Hide Explanation

The correct answer is increasing inotropy (A). The main clinical inotropes are digoxin, dobutamine, and milrinone. angiotensin-converting-enzyme (ACE) inhibitors and angiotensin-receptor blockers (ARBs) are excellent for treating heart failure for all the other listed reasons: preventing cardiac remodeling (B), preventing fluid retention (C), reducing afterload (D), and reducing preload (E).

2. Which of the following best describes the mechanism of action of digoxin and subsequent levels of intracellular Na^+ and Ca^{2+}?

A. ● $\downarrow Na^+/K^+$-ATPase activity $\rightarrow \uparrow [Na^+] \rightarrow \uparrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter
B. ○ $\downarrow Na^+/K^+$-ATPase activity $\rightarrow \uparrow [Na^+] \rightarrow \downarrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter
C. ○ $\downarrow Na^+/K^+$-ATPase activity $\rightarrow \downarrow [Na^+] \rightarrow \uparrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter
D. ○ $\uparrow Na^+/K^+$-ATPase activity $\rightarrow \uparrow [Na^+] \rightarrow \uparrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter
E. ○ $\uparrow Na^+/K^+$-ATPase activity $\rightarrow \uparrow [Na^+] \rightarrow \downarrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter

Hide Explanation

The correct answer is $\downarrow Na^+/K^+$-ATPase activity $\rightarrow \uparrow [Na^+] \rightarrow \uparrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter (A). Higher intracellular Na^+ leads to increased Na^+/Ca^{2+} antiporter activity and thus higher levels of Ca^{2+} influx. On the other hand, $\downarrow Na^+/K^+$-ATPase activity $\rightarrow \uparrow [Na^+] \rightarrow \downarrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter (B) would lead to weaker contractility since intracellular Ca^{+2} is required for myocardial contraction. Reduced Na^+/K^+ activity leads to increased intracellular Na^+ because it is not actively being pumped out; therefore $\downarrow Na^+/K^+$-ATPase activity $\rightarrow \downarrow [Na^+] \rightarrow \uparrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter (C), is ruled out as a choice. Digoxin inhibits the activity of the Na^+/K^+-ATPase; so $\uparrow Na^+/K^+$-ATPase activity $\rightarrow \uparrow [Na^+] \rightarrow \uparrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter (D) and $\uparrow Na^+/K^+$-ATPase activity $\rightarrow \uparrow [Na^+] \rightarrow \downarrow Ca^{2+}$ influx via Na^+/Ca^{2+} antiporter (E) are also ruled out as choices.

3. Which of the following drugs is least useful in managing systolic heart failure?

A. ○ Carvedilol
B. ● Chlorthalidone
C. ○ Lisinopril
D. ○ Losartan
E. ○ Spironolactone

Hide Explanation

The correct answer is chlorthalidone (B). Loop diuretics are important in heart failure (HF) management by reducing preload, but thiazides like chlorthalidone are generally not potent enough to be routinely useful. Carvedilol (A) is a nonselective adrenergic inhibitor (inhibits both α- and β-receptors); β-blocker activity has been shown to reduce morbidity and mortality, as have ACE inhibitors like lisinopril (C), ARBs like losartan (D), and spironolactone (E).

4. Which of the following drugs do not improve mortality in patients with systolic heart failure?

A. ● Amlodipine
B. ○ Enalapril
C. ○ Metoprolol
D. ○ Spironolactone
E. ○ Valsartan

Hide Explanation

The correct answer is amlodipine (A). The drugs that reduce mortality also reduce ventricular remodeling, and these include ACE inhibitors such as enalapril (B), β-blockers such as metoprolol (C) and the aldosterone antagonist spironolactone (D), and ARBs such as valsartan (E). Amlodipine is a calcium channel vasodilator that reduces afterload, but is not known to reduce mortality in heart failure.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe chronic heart failure (CHF) and how the body responds.~~
02 ☑ ~~Outline the classes of drugs used to treat chronic systolic heart failure and their mechanisms of action.~~
03 ☑ ~~Outline a treatment plan for chronic heart failure, both systolic and diastolic.~~

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Objective Confidence: 3/3